

Chapter 2

Accessing the Command Line

The default shell for users in Red Hat Enterprise Linux is the GNU Bourne-Again Shell (bash).

Bash is an improved version of one of the most successful shells used on UNIX-like systems, the Bourne Shell (sh)

- When a regular user starts a shell, the default prompt **ends with a \$ character**
- The \$ character is replaced by a **# character** if the shell is running as the superuser, **root**.

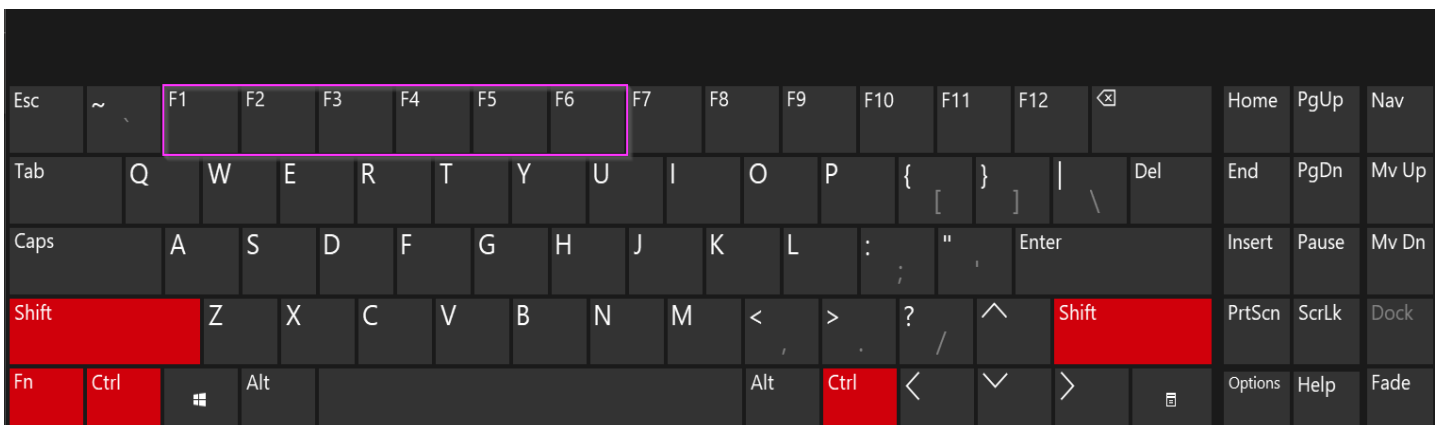
```
[user@host ~]$
```

```
[root@host ~]#
```

[root@host ~]#

- root: The logged-in username is root, the superuser account with full system privileges.
- @host: The @ separates username and hostname; “host” is the system’s hostname, often the default on fresh installs.
- ~ : The tilde represents the user’s home directory; for root this typically maps to /root, and it changes as directories change.
- #: The closing bracket ends the bracketed info, and # indicates a root shell prompt; regular users typically see \$ instead.

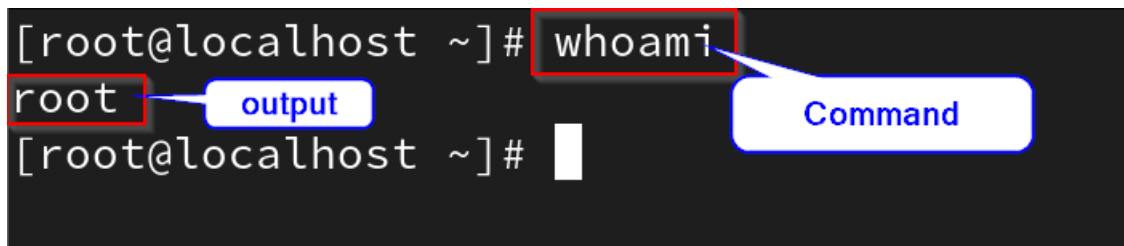
- This is the Linux machine's **physical console**.
- The physical console **supports multiple virtual consoles**, which can run separate terminals.
- Each virtual console supports an **independent login session**.
- You can switch between them by pressing **Ctrl + Alt** and a function key (**F1 through F6**) at the same time.
 - Where
 - F1 Will be Login Screen
 - F2 is 1st Gui Bash Terminal
 - F3 is CLI shell
 - F4 is CLI shell
 - F5 is CLI shell
 - F6 is CLI shell
- Most of these virtual consoles run a terminal providing a text login prompt, and if you enter your username and password correctly, you will log in and get a shell prompt.



❖ Executing Commands Using the Bash Shell

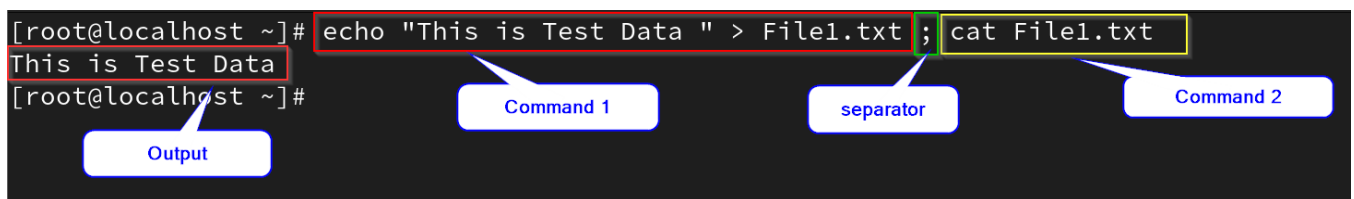
1 :- The **whoami** command prints the identity of the current user context

```
[root@localhost ~]# whoami
root
```



2 :- The following example shows how to combine two commands (**command1** and **command2**) on the command line use the **semicolon (;)** as a **command separator**

```
[root@localhost ~]# echo "This is Test Data " > File1.txt ; cat File1.txt
This is Test Data
```



3:- The date command displays the current date and time.

- date --- To Display Date and Time
- date +%R --- To Display Only Time
- date +%x --- To Display Only Date

```
[root@localhost ~]# date
Wednesday 27 August 2025 12:54:10 PM IST
[root@localhost ~]# date +%R
12:54
[root@localhost ~]# date +%x
27/08/25
[root@localhost ~]#
```

4 :- The **file** command scans the beginning of a file's **contents** and displays what **type** it is

```
[root@localhost ~]# file /etc/passwd
/etc/passwd: ASCII text
[root@localhost ~]# file /root
/root: directory
[root@localhost ~]# file /etc/yum.conf
/etc/yum.conf: symbolic link to dnf/dnf.conf
[root@localhost ~]#
```

5 :- To set date and time

```
[root@localhost ~]# date -s 12/27/2025 # date -s mm/dd/yyyy
Saturday 27 December 2025 12:00:00 AM IST
[root@localhost ~]# date -s 23:44:19 # date -s hrs:min:sec
Saturday 27 December 2025 11:44:19 PM IST
[root@localhost ~]#
```

To change Date

To Change Time

Format

6 :- To create directory

```
[root@localhost ~]# mkdir Folder1
[root@localhost ~]#
```

7 :- To check directory is created or not

```
[root@localhost ~]# mkdir Folder1
[root@localhost ~]# ls
anaconda-ks.cfg  Documents  Folder1  Pictures  Templates
Desktop          Downloads  Music    Public    Videos
[root@localhost ~]#
```

8 :- To create multiple directories

```
[root@localhost ~]# mkdir Folder2 Folder3 Folder4
[root@localhost ~]# ls
anaconda-ks.cfg  Downloads  Folder3  Pictures  Videos
Desktop          Folder1   Folder4  Public
Documents        Folder2   Music    Templates
[root@localhost ~]#
```

9 :- To create blank file

```
[root@localhost ~]# touch file1.txt
[root@localhost ~]# ls
anaconda-ks.cfg  Downloads  Folder2  Music  Templates
Desktop          file1.txt  Folder3  Pictures Videos
Documents        Folder1   Folder4  Public
```

10 :- To create a file with content

```
[root@localhost ~]# cat > file1.txt
This is File1
^Z
[1]+  Stopped                  cat > file1.txt
[root@localhost ~]#
```

Content

Command

Ctrl + Z to Save

11 :- .To check working directory

```
[root@localhost ~]# pwd
/root
[root@localhost ~]#
```

Print Working Dir

12 :- To long list of all files

```
[root@localhost ~]# ll
total 8
-rw-----. 1 root root 843 Mar 31 2025 anaconda-ks.cfg
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Desktop
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Documents
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Downloads
-rw-r--r--. 1 root root 14 Dec 28 00:04 file1.txt
drwxr-xr-x. 2 root root  6 Dec 27 23:58 Folder1
drwxr-xr-x. 2 root root  6 Dec 28 00:00 Folder2
drwxr-xr-x. 2 root root  6 Dec 28 00:00 Folder3
drwxr-xr-x. 2 root root  6 Dec 28 00:00 Folder4
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Music
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Pictures
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Public
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Templates
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Videos
```

```
[root@localhost ~]# ls -l
total 8
-rw-----. 1 root root 843 Mar 31 2025 anaconda-ks.cfg
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Desktop
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Documents
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Downloads
-rw-r--r--. 1 root root 14 Dec 28 00:04 file1.txt
drwxr-xr-x. 2 root root  6 Dec 27 23:58 Folder1
drwxr-xr-x. 2 root root  6 Dec 28 00:00 Folder2
drwxr-xr-x. 2 root root  6 Dec 28 00:00 Folder3
drwxr-xr-x. 2 root root  6 Dec 28 00:00 Folder4
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Music
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Pictures
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Public
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Templates
drwxr-xr-x. 2 root root  6 Aug 19 13:16 Videos
```

13 :- To change directory

```
[root@localhost ~]# cd Folder1
[root@localhost Folder1]#
```

Change Dir to Folder1

Changed Successfully

14 :- To create multiple folders with ID's

```
[root@localhost ~]# mkdir Folder{1..100}
mkdir: cannot create directory 'Folder1': File exists
mkdir: cannot create directory 'Folder2': File exists
mkdir: cannot create directory 'Folder3': File exists
mkdir: cannot create directory 'Folder4': File exists
[root@localhost ~]# ls
anaconda-ks.cfg  Folder14  Folder24  Folder34  Folder44  Folder54  Folder64  Folder74  Folder84  Folder94
Desktop          Folder15  Folder25  Folder35  Folder45  Folder55  Folder65  Folder75  Folder85  Folder95
Documents        Folder16  Folder26  Folder36  Folder46  Folder56  Folder66  Folder76  Folder86  Folder96
Downloads        Folder17  Folder27  Folder37  Folder47  Folder57  Folder67  Folder77  Folder87  Folder97
file1.txt        Folder18  Folder28  Folder38  Folder48  Folder58  Folder68  Folder78  Folder88  Folder98
Folder1          Folder19  Folder29  Folder39  Folder49  Folder59  Folder69  Folder79  Folder89  Folder99
Folder10         Folder2   Folder3   Folder4   Folder5   Folder6   Folder7   Folder8   Folder9   Music
Folder100        Folder20  Folder30  Folder40  Folder50  Folder60  Folder70  Folder80  Folder90  Pictures
Folder11         Folder21  Folder31  Folder41  Folder51  Folder61  Folder71  Folder81  Folder91  Public
Folder12         Folder22  Folder32  Folder42  Folder52  Folder62  Folder72  Folder82  Folder92  Templates
Folder13         Folder23  Folder33  Folder43  Folder53  Folder63  Folder73  Folder83  Folder93  Videos
[root@localhost ~]#
```

15 :- To create chain folder

```
[root@localhost ~]# mkdir F1/F2/F3/F4/F5/F6/F7
mkdir: cannot create directory 'F1/F2/F3/F4/F5/F6/F7': No such file or directory
[root@localhost ~]# mkdir -p F1/F2/F3/F4/F5/F6/F7
[root@localhost ~]#
```

16 :- To Delete Folders Created

```
[root@localhost ~]# rmdir Folder{1..100}
[root@localhost ~]# ls
anaconda-ks.cfg  Desktop  Documents  Downloads  F1  file1.txt  Music  Pictures  Public  Templates  Videos
[root@localhost ~]#
```

17 :- To create hidden Dir

```
[root@localhost ~]# mkdir .HiddenDir
[root@localhost ~]# ls
anaconda-ks.cfg  Desktop  Documents  Downloads  F1  file1.txt  Music  Pictures  Public  Templates  Videos
[root@localhost ~]# ls -a
.          .bash_history  .bashrc  .cshrc  Downloads  .HiddenDir  Pictures  .tcshrc
..         .bash_logout  .cache   Desktop  F1          .local      Public    Templates
anaconda-ks.cfg  .bash_profile .config  Documents file1.txt    Music      .ssh      Videos
[root@localhost ~]#
```

18.To view hidden file

```
[root@localhost ~]# ls -a
.          .bash_history  .bashrc  .cshrc  Downloads  .HiddenDir  Pictures  .tcshrc
..         .bash_logout  .cache   Desktop  F1          .local      Public    Templates
anaconda-ks.cfg  .bash_profile .config  Documents file1.txt    Music      .ssh      Videos
[root@localhost ~]#
```

19 :- To hide existing folder

```
[root@localhost ~]# ls
anaconda-ks.cfg Desktop Documents Downloads file1.txt Music Pictures Public Templates Videos
[root@localhost ~]# mv F1 .F1
[root@localhost ~]# ls
anaconda-ks.cfg Desktop Documents Downloads file1.txt Music Pictures Public Templates Videos
[root@localhost ~]# ls -a
.          .bash_history .bashrc .cshrc Downloads .HiddenDir Pictures .tcshrc
..         .bash_logout .cache Desktop .F1         .local   Public  Templates
anaconda-ks.cfg .bash_profile .config Documents file1.txt Music   .ssh    Videos
[root@localhost ~]#
```

20 :- To edit existing file

```
[root@localhost ~]# cat file1.txt
This is File1
[root@localhost ~]# cat >> file1.txt
New Data Test
^Z
[2]+  Stopped                  cat >> file1.txt
[root@localhost ~]# cat file1.txt
This is File1
New Data Test
[root@localhost ~]#
```

21.To view edited file content

```
[root@localhost ~]# cat file1.txt
This is File1
New Data Test
[root@localhost ~]#
```

22.To delete individual file or folder

```
[root@localhost ~]# ls
anaconda-ks.cfg Desktop Documents Downloads file1.txt Music Pictures Public Templates Videos
[root@localhost ~]# rm -rf file1.txt
[root@localhost ~]# rm -rf Desktop/
[root@localhost ~]# ls
anaconda-ks.cfg Documents Downloads Music Pictures Public Templates Videos
[root@localhost ~]#
```

23 :- Using Man Command

```
[root@localhost ~]# man mkdir
[root@localhost ~]#
```

```
MKDIR(1) User Commands

NAME
    mkdir - make directories

SYNOPSIS
    mkdir [OPTION]... DIRECTORY...

DESCRIPTION
    Create the DIRECTORY(ies), if they do not already exist.

    Mandatory arguments to long options are mandatory for short options too.

    -m, --mode=MODE
        set file mode (as in chmod), not a=rwx - umask

    -p, --parents
        no error if existing, make parent directories as needed

    -v, --verbose
        print a message for each created directory

Manual page mkdir(1) line 1 (press h for help or q to quit)
```

24 :- To rename a file

```
[root@localhost ~]# ls
anaconda-ks.cfg Documents Downloads Music Pictures Public Templates Videos
[root@localhost ~]# mv anaconda-ks.cfg demo-file.txt
[root@localhost ~]# ls
demo-file.txt Documents Downloads Music Pictures Public Templates Videos
[root@localhost ~]#
```

25.To copy content of one file to another

```
[root@localhost ~]# ls
demo-file.txt Documents Downloads file1.txt Music Pictures Public Templates Videos
[root@localhost ~]# cat file1.txt
This is File1
[root@localhost ~]# touch file2.txt
[root@localhost ~]# cat file2.txt
[root@localhost ~]# cp file1.txt file2.txt
cp: overwrite 'file2.txt'? yes
[root@localhost ~]# cat file2.txt
This is File1
[root@localhost ~]#
```


26.To view calendar of current month

```
[root@localhost ~]# cal
    December 2025
Su Mo Tu We Th Fr Sa
  1  2  3  4  5  6
  7  8  9 10 11 12 13
14 15 16 17 18 19 20
21 22 23 24 25 26 27
28 29 30 31
[root@localhost ~]#
```

27.To view calendar of particular month

```
[root@localhost ~]# cal 05 2025
    May 2025
Su Mo Tu We Th Fr Sa
          1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31
[root@localhost ~]#
```

28. To view calendar of whole year

```
[root@localhost ~]# cal 2025
                2025

    January      February      March
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
  1  2  3  4          1          1
 5  6  7  8  9 10 11  2  3  4  5  6  7  8  2  3  4  5  6  7  8
12 13 14 15 16 17 18  9 10 11 12 13 14 15  9 10 11 12 13 14 15
19 20 21 22 23 24 25 16 17 18 19 20 21 22 16 17 18 19 20 21 22
26 27 28 29 30 31    23 24 25 26 27 28    23 24 25 26 27 28 29
                                30 31

    April        May          June
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
  1  2  3  4  5          1  2  3          1  2  3  4  5  6  7
 6  7  8  9 10 11 12  4  5  6  7  8  9 10  8  9 10 11 12 13 14
13 14 15 16 17 18 19 11 12 13 14 15 16 17 15 16 17 18 19 20 21
20 21 22 23 24 25 26 18 19 20 21 22 23 24 22 23 24 25 26 27 28
27 28 29 30    25 26 27 28 29 30 31 29 30

    July         August       September
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
  1  2  3  4  5          1  2          1  2  3  4  5  6
 6  7  8  9 10 11 12  3  4  5  6  7  8  9  7  8  9 10 11 12 13
13 14 15 16 17 18 19 10 11 12 13 14 15 16 14 15 16 17 18 19 20
20 21 22 23 24 25 26 17 18 19 20 21 22 23 21 22 23 24 25 26 27
27 28 29 30 31    24 25 26 27 28 29 30 28 29 30

    October      November     December
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
  1  2  3  4          1          1  2  3  4  5  6
 5  6  7  8  9 10 11  2  3  4  5  6  7  8  7  8  9 10 11 12 13
12 13 14 15 16 17 18  9 10 11 12 13 14 15 14 15 16 17 18 19 20
19 20 21 22 23 24 25 16 17 18 19 20 21 22 21 22 23 24 25 26 27
26 27 28 29 30 31    23 24 25 26 27 28 29 28 29 30 31

                                30
[root@localhost ~]#
```

29.To view binary calculator

```
[root@localhost ~]# bc #binary calculator
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
234+567
801
123*456
56088
123/456
0
123-21
102
^Z
[2]+ Stopped          bc
[root@localhost ~]#
```

30.To power off the system

```
[root@localhost ~]# poweroff
```

Or

```
[root@localhost ~]# init 0
```

31.To restart a system

```
[root@localhost ~]# reboot
```

Or

```
[root@localhost ~]# init 6
```

32.To create a shortcut commands

```
[root@localhost ~]# alias c=clear
[root@localhost ~]# alias m=mkdir
[root@localhost ~]# m Folder1
[root@localhost ~]# c
```

33.To view top lines of the file

```
[root@localhost ~]# head /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
[root@localhost ~]#
```

37.To view bottom lines of the file

```
[root@localhost ~]# tail /etc/passwd
colord:x:986:986:User for colord:/var/lib/colord:/sbin/nologin
setroubleshoot:x:985:985:SELinux troubleshoot server:/var/lib/setroubleshoot:/usr/sbin/nologin
clevis:x:984:984:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/usr/sbin/nologin
gdm:x:42:42::/var/lib/gdm:/sbin/nologin
gnome-initial-setup:x:983:983::/run/gnome-initial-setup:/sbin/nologin
chrony:x:982:982:chrony system user:/var/lib/chrony:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/usr/sbin/nologin
dnsmasq:x:981:981:Dnsmasq DHCP and DNS server:/var/lib/dnsmasq:/usr/sbin/nologin
tcpdump:x:72:72::/sbin/nologin
student:x:1000:1000:student:/home/student:/bin/bash
[root@localhost ~]#
```

38. To view history of previous commands

```
[root@localhost ~]# history
```

39.To run particular command from history

40.To clear history

```
[root@localhost ~]# history -c
[root@localhost ~]# history
1 history
[root@localhost ~]#
```

```
95 c
96 alias c=clear
97 alias m=mkdir
98 m Folder1
99 head /etc/passwd
100 tail /etc/passwd
101 history
[root@localhost ~]# !99
```

Useful Command-line Editing Shortcuts

Shortcut	Description
Ctrl+A	Jump to the beginning of the command line.
Ctrl+E	Jump to the end of the command line.
Ctrl+U	Clear from the cursor to the beginning of the command line.
Ctrl+K	Clear from the cursor to the end of the command line.
Ctrl+LeftArrow	Jump to the beginning of the previous word on the command line.
Ctrl+RightArrow	Jump to the end of the next word on the command line.
Ctrl+R	Search the history list of commands for a pattern.